

A Combined ODE Model of Carbohydrate Fermentation and Colorectal Cancer

A. Lawryshyn, H.J. Eberl, T. Hillen

Abstract

We formulate and analyze a system of non-linear ordinary differential equations that describe key metabolic and immunological interactions between butyrate produced by fiber-fermenting gut microbiota, colorectal cancer cells and host cell populations. The model is studied both independently and in conjunction with a pre-existing carbohydrate fermentation model. The parameter space is explored through sensitivity analyses. Simulation experiments are conducted to illustrate the emergence of varying dynamical behaviour driven by butyrate availability. Our model predicts that butyrate production is driven by fiber consumption and further supported by probiotics in the case of microbial dysbiosis. It also suggests that butyrate may help in suppressing tumour growth. We also show that by adding noise with sufficiently high intensity, cancer elimination occurs almost surely in infinite time and that this threshold level of noise intensity decreases with increasing butyrate concentrations.

1 Introduction

1.1 The Human Colon and Gut Microbiome

The human colon is host to a diverse population of microorganisms, including bacteria, fungi, and archaea, that are collectively referred to as gut microbiota. These microbiota play an essential role in digestion, metabolism, and immune system support. The gut microbiome has become a popular research topic over the past few decades, with increasing evidence highlighting its connection to a wide range of diseases including inflammatory bowel diseases (IBD), irritable bowel syndrome (IBS), obesity, diabetes and cancer [1, 2]. The specific composition of the gut microbiome is unique to each individual, depending on factors such as genetics, diet, and antibiotic use [1].

Fiber fermenting bacteria, including species from the *Bacterioides* and *Bifidobacterium* genera in addition to others [3], are responsible for the fermentation of complex carbohydrates into short-chain fatty acids (SCFAs). Acetate, propionate and butyrate are the most abundant SCFAs in the gut. They are essential to human health, playing a significant role in energy metabolism, gut barrier strength and immune response modulation [4].

1.1.1 Carbohydrate Fermentation

The human colon is lined with a two-layer mucus membrane: an inner layer that adheres to the epithelial cells and is devoid of bacteria, and an outer, unattached layer which is colonized by microbiota [5]. The lumen of the colon is also colonized but encompasses a different microbial composition than that of the mucus layer.

Glycans (dietary fiber and mucins) are the main energy sources for carbohydrate fermenting gut microbiota. The majority of dietary fibers originate from plant cell walls, obtained by the consumption of fruits, vegetables and whole grains [6]. Dietary fiber escapes digestion in the upper gastrointestinal (GI) tract and enters the lumen of the colon through the small intestine. Conversely, mucins, which are glycoproteins produced by the host’s epithelial tissue, are abundant in the mucus layer. Only certain microbial species are capable of degrading fiber and mucins into simpler sugars. These sugars are further broken down by other species into SCFAs, lactate, and gas.

1.2 The Gut Microbiome and Colorectal Cancer

Colorectal cancer (CRC) is the third most common type of cancer and the second leading cause of cancer-related deaths worldwide [7]. Growing evidence highlights important connections between CRC and the gut microbiome [2, 8, 9, 10]. In particular, it has been shown that butyrate, produced from fiber-fermenters in the gut, has an inhibitory effect on cancer cell growth but promotes that of healthy cells. This conflicting effect of butyrate, known as the “butyrate paradox”, is thought to be related to the contrasting metabolic behaviour of normal versus cancerous cells as well as the supportive role of butyrate on the immune system [11, 12].

1.2.1 Metabolic role of butyrate in cancer growth

Butyrate is present in high concentrations in the human colon, and is utilized by colonocytes as their primary energy source, accounting for approximately 70% of their total energy requirements [11]. Unlike colonocytes, which preferentially metabolize butyrate via beta-oxidation, cancer cells undergo the Warburg effect [13], relying on glucose as an energy source, which favours their fast growth and survival [14]. Butyrate has been shown to inhibit this effect by acting as a histone deacetylase (HDAC) inhibitor, shifting cancer cell metabolism toward respiration and thereby slowing proliferation [11].

1.2.2 Butyrate affects immune response to cancer

The immune system’s response to cancer is highly complex, involving many different biochemical pathways. We focus on a high level overview, incorporating three key types of immune cells in our model - cytotoxic T cells, regulatory T cells (Tregs) and dendritic cells.

Dendritic cells capture tumour antigens and present them to T cells, causing their activation and recruitment to the tumour microenvironment, initiating an immune response [15].

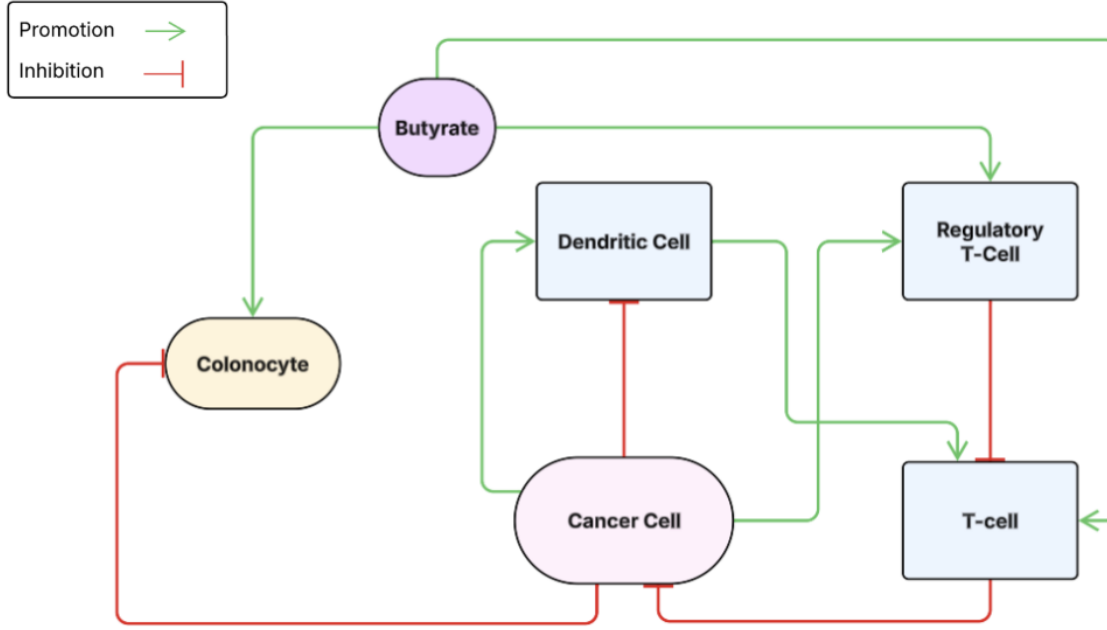


Figure 1: Diagram of primary interactions of butyrate, cancer and normal cells with the immune system.

T cells can kill the cancer cells by releasing molecules that induce apoptosis [16]. A high presence of T cells is often associated with better control of cancer progression, although certain tumours can evade destruction by impairing T cell infiltration and function [17]. Tregs are important for preventing autoimmunity but in the presence of cancer, they can directly inhibit T-cells, creating an immunosuppressive environment that allows for cancer growth [18].

Through its role as a HDAC inhibitor, butyrate increases the expression of the protein Foxp3 which is the primary transcription factor required for Treg development and function [19]. Butyrate also boosts T cell function and proliferation [20]. Figure 1 shows a visual depiction of the key stimulatory and inhibitory interactions between butyrate, cancer cells, colonocytes, and immune cells.

1.3 Effect of Probiotic Administration on Butyrate Production and CRC Progression

1.4 Role of Fiber in CRC Immunotherapy

Immunotherapy is a targeted treatment that helps strengthen the immune system's response to cancer. It has transformed treatment strategies against many types of cancers but immunotherapy efficacy against CRC is currently limited [21]. Tumours with high microsatellite

instability (MSI) are often treated with Programmed Cell Death Protein 1 (PD-1) inhibitors. PD-1 is a protein on T cells that, under normal conditions, help prevent them from attacking other cells in the body. However, only 10%-15% of total CRC cases have high MSI, with the rest being microsatellite stable (MSS) [22]. Microsatellites are short, repetitive sequences of DNA distributed throughout the genome. These regions are particularly susceptible to errors during DNA replication, most of which are corrected by the DNA mismatch repair (MMR) system. However, when the MMR system is impaired, these errors accumulate and can lead to alterations in microsatellite sequences, causing MSI. Butyrate has been shown to improve efficacy of PD-1 inhibitors in both MSI-high and MSS tumours, suggesting that increased fiber intake may be beneficial for CRC patients who are resistant to PD-1 inhibitors [20, 23].

1.5 Cancer Modelling

Mathematical oncology has gained traction over the past several decades, providing a theoretical framework for clinical studies [24]. In-silico experiments are now prevalent in the medical field, as in-vitro and in-vivo experiments are often expensive, time-consuming, or ethically challenging. In this thesis, we construct an ordinary differential equation (ODE) model describing interactions between butyrate produced by gut microbiota and colon cancer. Several CRC models have been developed that incorporate gut dynamics. The authors in [25] constructed a compartmental model of CRC cell population dynamics in the colonic crypt, describing interactions between stem cells, differentiated cells and transit cells. In [26], a reaction-diffusion model of colitis-associated colon cancer was formulated to study the impact of chronic inflammation of the colon mucosa on tumour suppressing genes. A model of CRC initiation in the colonic crypt was introduced in [27] and a reaction-diffusion system of colon cancer cell metabolic reprogramming was studied in [28].

Our model is constructed to focus on cancer-butyrate dynamics, which primarily involves interactions with specific immune cells — T cells, Tregs and dendritic cells, but many cancer models have been developed that include different combinations of immune elements. For example, [29] and [30] formulate normal and cancer growth as competition models, both including immune cells as a single component. The model in [31] includes colon cancer, dendritic, helper T and cytotoxic T cells. Additionally, [32] and [33] constructed detailed models for breast and colon cancer respectively, each incorporating various types of immune cells (including T cells, Tregs and dendritic cells) and cytokines. In [34], a large ODE system was constructed that incorporates interactions between colon cancer cells, immune cells, and various families of gut microbiota. A comprehensive review of cancer-immune cell modelling is presented in [35]. Butyrate was incorporated in an immunological model describing inflammation in the human gut [36], as well as in a model of the gut-bone axis [37]. However, to the best of our knowledge, no existing mathematical models describe the role of butyrate as a cellular metabolite in CRC.